

WORK SHEET - 04

VECTOR & 3D

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.6 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 Let L be the line passing through the point P (1, 2, 0) and perpendicular to the plane $\vec{r} \cdot (3\hat{i} + 4\hat{k}) = 0$.
If line L intersects the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 13$ at point Q, then co-ordinates of Q equals
(A) (3, -6, 4) (B) (-2, -4, 11) (C) (7, 2, 8) (D) (4, 0, 9)
- Q.2 The volume of parallelopiped with its edges represented by the vectors $\hat{i} + \pi\hat{k}$, $3\hat{i} + \pi\hat{k}$ and $-4\hat{i} + 2\hat{j} + \pi\hat{k}$ is equal to
(A) π (B) 2π (C) 3π (D) 4π
- Q.3 Let P be the plane passing through the point (0, 1, -1) and perpendicular to the line of intersection of the planes $2x + z = 3$ and $3y + 2z = 5$. If plane P intersects x-axis, y-axis, z-axis at points A, B, C respectively then area of triangle ABC equals
(A) $\frac{13\sqrt{61}}{36}$ (B) $\frac{11\sqrt{61}}{36}$ (C) $\frac{25\sqrt{61}}{36}$ (D) $\frac{8\sqrt{61}}{36}$
- Q.4 If $\vec{v}_1 = 8\hat{i} + \hat{j} + (a^2 + 4a + 5)\hat{k}$, $\vec{v}_2 = -4\hat{i} + \hat{j} + 4\cot^2 \theta \hat{k}$ and $\vec{v}_3 = -5\hat{i} - \hat{j} + \tan^2 \theta \hat{k}$, where $a \in \mathbb{R}$ and $\theta \in \mathbb{R} - \left\{ n\pi, (2n+1)\frac{\pi}{2}, n \in \mathbb{I} \right\}$, represents three coterminal edges of a tetrahedron then minimum volume of the tetrahedron in cubic units, is
(A) 31 (B) 33 (C) $\frac{11}{2}$ (D) $\frac{7}{3}$
- Q.5 The least value of the volume of parallelepiped formed by the vectors $\vec{V}_1 = \hat{i} + \hat{j}$, $\vec{V}_2 = \hat{i} + (2\operatorname{cosec} \alpha)\hat{j} + \hat{k}$ and $\vec{V}_3 = \hat{j} + (2\operatorname{cosec} \alpha)\hat{k}$ where $\alpha \in (0, \pi)$, is
(A) 16 (B) 9 (C) 5 (D) 1
- Q.6 Equation $\frac{x}{1} = \frac{y-2}{3} = \frac{z+5}{2}$ and $\frac{x-1}{2} = \frac{y-\lambda}{6} = \frac{z-\mu}{4}$ represent same line then $(\lambda - \mu)$ is equal to
(A) 7 (B) 5 (C) 6 (D) 8

[PARAGRAPH TYPE]

Q.7 to Q.9 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 7 to 9

Let $L_1 : \vec{r} = (7\hat{i} + 6\hat{j} + 2\hat{k}) + \lambda(-3\hat{i} + 2\hat{j} + 4\hat{k})$ and $L_2 : \vec{r} = (5\hat{i} + 3\hat{j} + 4\hat{k}) + \mu(2\hat{i} + \hat{j} + 3\hat{k})$ where $\lambda, \mu \in \mathbb{R}$ be two lines in space. A line L_3 parallel to vector $\vec{v} = 2\hat{i} - 2\hat{j} - \hat{k}$ intersects L_1 and L_2 at A and B respectively.

- Q.7 The length AB is equal to
(A) 4 (B) 6 (C) 9 (D) 10
- Q.8 The equation of plane parallel to line L_1 and containing line L_2 is equal to
(A) $x + 3y + 4z = 30$ (B) $x + 2y + z = 15$ (C) $2x - y + z = 11$ (D) $2x + 17y - 7z = 33$
- Q.9 The volume of tetrahedron OPAB (where O is origin) and P(1, 2, 3) is equal to
(A) 5 (B) 7 (C) 9 (D) 10

[MULTIPLE CORRECT CHOICE TYPE]

Q.10 & Q.11 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct. **[2 × 4 = 8]**

- Q.10 If $\vec{a}, \vec{b}$ are non-parallel unit vectors and $\vec{c}$ is a vector in a plane perpendicular to $\vec{a}$ and $\vec{b}$ such that $|\vec{c}| = 5$ and $|\vec{a} + \vec{b}|^2 = |\vec{a} \times \vec{b}|$ then
(A) $\vec{a} \cdot \vec{b} = \frac{3}{5}$ (B) $\vec{a} \cdot \vec{b} = \frac{-3}{5}$ (C) $[\vec{a} \vec{b} \vec{c}] = -4$ (D) $[\vec{a} \vec{b} \vec{c}] = 4$
- Q.11 Consider two planes $P_1 : 2x - 3y + z = -5$ and $P_2 : x - 4y - 14z = 5$. If a line L which is perpendicular to line of intersection of planes P_1 & P_2 and whose one point $A(-7, y_1, z_1)$ lies on the plane P_1 and points $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ lie on plane P_2 has the equation $\frac{x+7}{a} = \frac{y-y_1}{-b} = \frac{z-z_1}{c}$, where $a, b, c \in \mathbb{N}$, then
(A) least value of $(a + b + c)$ is 6. (B) $y_1 + z_1$ is equal to -3.
(C) $\frac{z_3 - z_2}{y_3 - y_2} = 3$ (D) $\frac{x_3 - x_2}{z_3 - z_2} = 2$

[MATRIX TYPE]

Q.12 & Q.13 has **four** statements (A, B, C, D) given in **Column-I** and **five** statements (P, Q, R, S, T) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.
[2 (3 + 3 + 3 + 3) = 24]

Q.12 Consider a matrix, $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$ where $a_i < a_{i+1} \forall i = 1, 2, \dots, 8$ and a_i 's are the divisors of

a smallest natural number N.

Column-I	Column-II
(A) The absolute value of $\det(A)$ equals	(P) 13
(B) The value of $\text{Tr}(A)$ equals	(Q) 18
(C) If $\vec{v}_1 = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{v}_2 = a_4\hat{i} + a_5\hat{j} + a_6\hat{k}$ then $\vec{v}_1 \cdot \vec{v}_2$ is equal to	(R) 36
(D) The value of $\int_{a_1}^{a_6} [\sqrt{x}] dx$ is equal to	(S) 43
	(T) 49

Note : $\text{Tr}(P)$ and $\det(P)$ denote trace of matrix P and determinant of matrix P respectively and $[k]$ denotes the greatest integral value less than or equal to k.

Q.13 **Column-I** **Column-II**

(A) If $\vec{p} \cdot (\vec{q} + \vec{r}) + \vec{q} \cdot (\vec{r} + \vec{p}) + \vec{r} \cdot (\vec{p} + \vec{q}) = 0$ and $ \vec{p} = 1, \vec{q} = 4, \vec{r} = 8$ then $ \vec{p} + \vec{q} + \vec{r} $ is equal to	(P) 2
(B) Let A be a point on the line $\frac{x+3}{2} = \frac{y-6}{3} = \frac{z-3}{-2}$ and B be a point on the line $\frac{x-0}{2} = \frac{y-6}{2} = \frac{z-0}{-1}$. The least distance AB is equal to	(Q) 3
(C) A point (α, β, γ) lies on the plane $x + 2y + z = 4$. The value of β such that the vector $\vec{v} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ satisfies $\hat{j} \times (\hat{j} \times \vec{v}) = \vec{0}$, is equal to	(R) 4
(D) A variable plane at a distance of 2 unit from the origin cuts the co-ordinate axes at A, B and C. If the centroid D (x, y, z) of tetrahedron OABC satisfies the relation $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = k$, then the value of k is	(S) 6 (T) 9

[INTEGER TYPE]

Q.14 to Q.18 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

- Q.14 A point P moves in a plane $\Pi : 2x - 3y + 6z - 4 = 0$ such that area of ΔPAB where $A \equiv (2, 2, 1)$ and $B \equiv (-1, -4, -1)$ is always equal to 14 sq. units. If the plane perpendicular to the plane Π containing the locus of P are $6x + ay + bz + d_1 = 0$ and $6x + ay + bz + d_2 = 0$, $d_1 > d_2$ then find the value of $\left(\frac{d_1 - d_2 + a + b}{17} \right)$.
- Q.15 Two intersecting lines $\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k} + \lambda(\hat{i} + \hat{j})$ and $\vec{r} = \hat{i} + 4\hat{j} + 5\hat{k} + \mu(2\hat{j} + \hat{k})$ are lying in a plane Π_1 . If the equation of plane Π_2 is $x - y + 2z - 19 = 0$ and distance between the planes Π_1 and Π_2 is d, then find the value of d^2 .
- Q.16 A plane P passes through the intersection of planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) + 3 = 0$ and $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 2$. If plane P divide the line segment joining $M(3, 0, 2)$ and $N(0, 3, -1)$ in 2 : 1 (internally) and the equation of plane P is $\vec{r} \cdot (a\hat{i} - \hat{j} + b\hat{k}) = c$ where $a, b, c \in \mathbb{N}$, then find $(a + b + c)$.
- Q.17 If the equation of the plane through the straight line $\frac{x-1}{2} = \frac{y+2}{-3} = \frac{z}{5}$ and perpendicular to the plane $x - y + z + 2 = 0$ is $ax - by + cz + 4 = 0$, then find the value of $(10^3a + 10^2b + 10c)$.
- Q.18 Let $\vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$ where $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero and non-coplanar vectors. If $\vec{r}$ is perpendicular to $(\vec{a} + \vec{b} + \vec{c})$, then minimum value of $(x^2 + y^2)$ is α . Value of least integer greater than or equal to $\tan^{-1} \left(\tan \frac{\alpha}{\pi} \right)$ is

[ANSWER KEY]

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|------------------------------------|------------------------------------|--------|-----------|-----------|
| Q.1 C | Q.2 D | Q.3 C | Q.4 C | Q.5 D |
| Q.6 D | Q.7 C | Q.8 D | Q.9 B | Q.10 BCD |
| Q.11 ABD | | | | |
| Q.12 (A) Q ; (B) S ; (C) S ; (D) P | Q.13 (A) T ; (B) Q ; (C) P ; (D) R | | | |
| Q.14 3 | Q.15 0024 | Q.16 7 | Q.17 1710 | Q.18 0001 |

